

Pre-Exam Installation List

Step 1 — Verify Python

```
python --version    # must be 3.10 or above
```

```
pip --version
```

If Python is not installed, download from **python.org** and add to PATH during installation.

Step 2 — Install All Python Packages

Run this single command:

```
pip install gradio openai opencv-python Pillow openai-whisper SpeechRecognition PyAudio  
sounddevice soundfile pydub numpy requests python-dotenv
```

If **PyAudio fails on Windows**, run these instead:

```
pip install pipwin
```

```
pipwin install pyaudio
```

Step 3 — Install ffmpeg (system-level, NOT pip)

Required for MP3 audio support in pydub.

Windows:

```
winget install ffmpeg
```

Or download manually from **ffmpeg.org** → add the bin folder to system PATH.

Verify:

```
ffmpeg -version
```

Step 4 — Install Ollama

1. Go to **ollama.com** and download the installer for your OS
2. Run the installer
3. After install, open a terminal and verify:

```
ollama --version
```

Step 5 — Download the LLM Model (4.7 GB — do this at home, not on exam day)

```
ollama pull llama3
```

Verify it works:

```
ollama run llama3 "Reply with one word: hello"
```

Type /bye to exit.

Step 6 — Pre-download the Whisper Model (~140 MB)

```
python -c "import whisper; whisper.load_model('base'); print('Whisper model ready')"
```

Step 7 — Final Verification (run the night before exam)

```
python -c "import gradio, cv2, whisper, speech_recognition, openai, PIL, pydub, numpy, requests; print('All packages OK')"
```

```
ollama run llama3 "Say: I am ready"
```

```
ffmpeg -version
```

All three must succeed. If any fail, fix before coming to exam.

Important Notes

openai package	Installed locally: used to talk to Ollama on your own laptop , NOT OpenAI's servers. No internet needed during exam.
Internet on exam day	Not available. All models and packages must be installed before you arrive.
Ollama during exam	Must be running before you launch app.py. Open a terminal, run ollama serve, leave it open.
Camera & Mic	Test that your laptop camera and microphone work before exam day.
Submission	All code in one folder named RegNo_Name. Run python app.py before time is called.